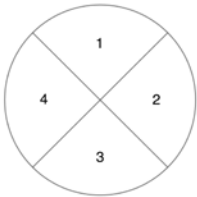


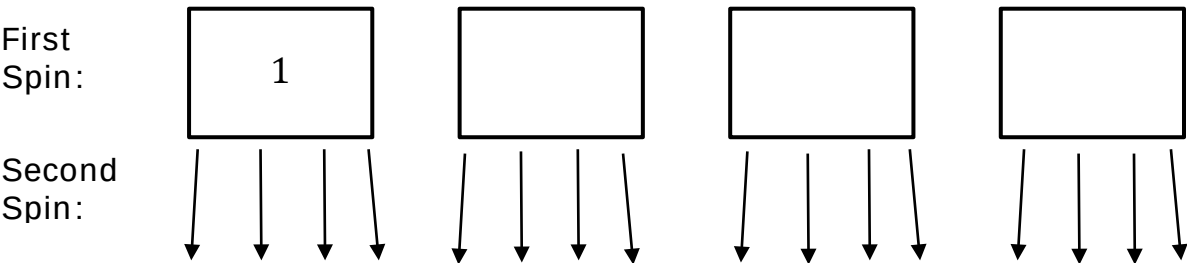
Name _____
Date _____
Period _____

1. List the sample space of picking a card out of the pile of hearts, replacing the first card after it's been picked.
2. Use the sample space to determine how likely it is that Brielle will choose the Queen of hearts both times. Create a ratio to compare the number of outcomes with only the Queen of Hearts to the total number of outcomes in the sample space.
3. Predict how the sample space and the probability of choosing the Queen of hearts each time would change if Brielle uses the whole deck of cards and not just the hearts.

Suppose a fair spinner numbered 1 through 4 is spun 2 times.
 Complete the tree diagram showing all possible outcomes of spinning the spinner twice. On the first spin, the options are 1,2,3, and 4. On the second spin, the options are 1,2,3, and 4.



4. Write the possible outcomes for the first spin.
5. Write the possible outcomes for the second spin.
6. Write the sample space.



Sample Space: { }

7. Complete the table.

Experiment	Number of Outcomes in the First Step	Number of Outcomes in the Second Step	Total Number of Outcomes in the Sample Space
Spinning a fair spinner (numbered 1 – 4) twice			

8. Create a tree diagram to determine the sample space of flipping a fair coin four times.
9. How many events are in Event A = Flipping all tails?
10. How many events are in Event B = Flipping exactly one tail?